Po-Yun Cheng

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Research Interests: My research goal is to make it cheaper for developers to verify AI-generated code, without letting software quality erode. I'm interested in designing interfaces, tools, and metrics to support this, combining methods from HCI, software engineering, and AI systems.

Education

Sep. 2022–Jun. 2026 **B.S. in Computer Science and Information Engineering** (GPA 3.5/4.0), *National Taiwan University*, Taipei, Taiwan

- Exchange student, *University of California*, *Merced* (Fall 2025).
- Relevant courses: Data Structure and Algorithm, Algorithm Design and Analysis, Computer Network, Machine Learning, Software Engineering*, Deep Learning*, Advanced Human-Computer Interaction*. (*graduate level)

Research Experience

Feb. 2025–Jul. 2026 **Undergraduate Research Assistant**, Advisor: Prof. Jonathan Lee
Software Engineering Lab, National Taiwan University (remote)

- Led a three-person team on automated code review; presented a poster at Taiwan's flagship software engineering conference (TCSE 2026), winning the Best Poster Paper Award.
- Designed a stage-based multi-agent code review framework combining static analysis and project-context RAG; evaluated against CodeRabbit, GitHub Copilot, and a single-LLM ablation on real open-source project PRs, achieving the highest F1 on review-comment similarity at 90% lower inference cost ($0.02 vs. $0.50 per review).
- Diagnosed a vocabulary-mismatch failure in embedding-based Java retrieval and fixed it via AST-derived summary enrichment, turning zero relevant retrievals into consistent correct ones.

Apr. 2024–Oct. 2024 **Undergraduate Research Assistant**, Advisor: Prof. Mike Y. Chen
Human-Computer Interaction Lab, National Taiwan University

- Designed the formative study and questionnaire (n=16) comparing VR headset use standing vs. lying down; found supine posture reduced head rotation range by 50% horizontally and 45% vertically, defining the actuation design and target ranges.
- Built the Python analysis pipeline, applying Wilcoxon signed-rank tests and Generalized Linear Mixed Models to repeated-measures data.
- Analyzed Likert-scale outcomes (comfort, preference, dizziness) for the summative evaluation and produced all statistical figures for the resulting CHI '25 paper.

Publications

1. Wu, E.-H., **Cheng, Po-Yun**, Hsu, C.-W., Han, C. H., Lee, P. C., Fan, C.-A., Kuo, Y. C., Hu, K.-J., Chen, Y. & Chen, M. Y. *HeadTurner: Enhancing Viewing Range and Comfort of using Virtual and Mixed-Reality Headsets while Lying Down via Assisted Shoulder and Head Actuation* in *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, 2025). `https://doi.org/10.1145/3706598.3714214`.

Presentations

1. **Cheng, Po-Yun**, Lee, K.-Y., Zhang, C.-W. & Lee, J. *Automated Code Review Using a Multi-LLMs Framework* Poster presented at the 22nd Taiwan Conference on Software Engineering (TCSE 2026). **Best Poster Paper Award**. National Chin-Yi University of Technology, Taichung, Taiwan, July 2026. `https://tcse2026.seat.org.tw/tcse2026-recap`.

Work Experience

Feb. 2026–Present

AI Engineer Intern,
AI Solution Center, Cathay Life Insurance

- Evaluated the performance (latency, memory, accuracy) of 8 quantization settings on local LLM inference; established a selection methodology adopted across departments for model deployment decisions.
- Built an LLM-based retrieval and clustering pipeline that surfaces insurance industry trends from 46 online sources, reducing analysts' monthly review time by 60%.

Projects

Fall 2025

Gesture-Based Pointer Control System and Usability Evaluation
Advanced Human-Computer Interaction course research project [github]

- Built an end-to-end touchless pointer system that maps palm location to cursor position and thumb–finger pinch dynamics to click and drag events.
- Designed three tasks (drag-drop, menu-select, and keyboard-input) to simulate the most frequent desktop interactions.
- Ran a counterbalanced within-subject study (Latin square, NASA-TLX + SUS), showing mid-air gesture input underperformed a touchpad and attributing this to inter-user gesture variability and sustained-elevation fatigue.

Fall 2025

Graph–Temporal Convolutional Network for Real-Time Gesture Recognition
Deep Learning course research project [github]

- Proposed GTCN, a causal graph-convolutional + dilated temporal-convolutional model for frame-wise dynamic hand gesture recognition on SHREC'21 dataset.
- Conducted a systematic ablation over 10 architectural variants (window length, spatial/temporal pooling, classifier design), isolating future context as a driver of temporal localization (Jaccard +19%) rather than detection accuracy.

Skills

Programming	C, Java (Spring), Python (PyTorch), TypeScript/JavaScript (React)
Tools	Git, Linux, Docker, Figma, LaTeX
Languages	Mandarin (native), English (TOEFL iBT 5.0/6.0; R=5.0, L=5.0, W=5.0, S=5.0)

Last updated: August 27, 2026